

Unit 11: The Derivative as a Rate of Change — Full Solutions

Warm-up

1. $s(t) = t^2 - 4t$

$$v(t) = s'(t) = 2t - 4 \quad a(t) = v'(t) = 2$$

$$v(1) = 2(1) - 4 = -2. \quad a(1) = 2.$$

Since $v(1) = -2$ is negative, the object is moving in the negative direction at $t = 1$.

2. $s(t) = 3t^2 + 2t - 5$

$$v(t) = 6t + 2 \quad a(t) = 6$$

$$v(0) = 6(0) + 2 = 2. \quad a(0) = 6.$$

3. $s(t) = t^3 - 3t^2$

$$v(t) = 3t^2 - 6t \quad a(t) = 6t - 6$$

$$v(2) = 3(4) - 6(2) = 12 - 12 = 0.$$

$v(2) = 0$ — the object is momentarily at rest at $t = 2$.

4. $s(t) = -16t^2 + 40t$

$$v(t) = -32t + 40$$

$v(0) = -32(0) + 40 = 40$. The initial velocity is 40 ft/s.

$$\text{Set } v(t) = 0: -32t + 40 = 0 \Rightarrow 32t = 40 \Rightarrow t = \frac{40}{32} = \frac{5}{4} = 1.25.$$

The object reaches its peak height at $t = 1.25$ seconds.

5. $s(t) = t^2 - 6t + 8$

$$v(t) = 2t - 6$$

$$v(1) = 2(1) - 6 = -4.$$

Since $v(1) = -4$ is negative, the object is moving in the negative direction at $t = 1$.

6. $s(t) = 2t^3 - 9t^2 + 12t$

$$v(t) = 6t^2 - 18t + 12 \quad a(t) = 12t - 18$$

$$v(1) = 6 - 18 + 12 = 0. \quad a(1) = 12 - 18 = -6.$$

Since $v(1) = 0$, the object is momentarily at rest at $t = 1$. ($a(1) = -6$.)

Standard

7. $s(t) = t^3 - 6t^2 + 9t$

$$v(t) = 3t^2 - 12t + 9 \quad a(t) = 6t - 12$$

At $t = 0.5$: $v(0.5) = 3(0.25) - 12(0.5) + 9 = 0.75 - 6 + 9 = 3.75$.

$$a(0.5) = 6(0.5) - 12 = 3 - 12 = -9.$$

Since $v(0.5) = 3.75 > 0$, the object is moving forward.

Since $v(0.5) > 0$ and $a(0.5) < 0$ have opposite signs, the object is slowing down.

8. $s(t) = t^2 - 8t + 7$

$$v(t) = 2t - 8 \quad a(t) = 2$$

At $t = 2$: $v(2) = 2(2) - 8 = -4$. Speed = $|v(2)| = 4$.

$$a(2) = 2 \text{ (constant, always 2).}$$

Since $v(2) = -4 < 0$, the object is moving in the negative direction, with speed 4.

Since $v(2) < 0$ and $a(2) > 0$ have opposite signs, the object is slowing down.

9. $s(t) = -t^3 + 3t^2$

$$v(t) = -3t^2 + 6t \quad a(t) = -6t + 6$$

At $t = 0.5$: $v(0.5) = -3(0.25) + 6(0.5) = -0.75 + 3 = 2.25$.

$$a(0.5) = -6(0.5) + 6 = -3 + 6 = 3.$$

Since $v(0.5) = 2.25 > 0$ and $a(0.5) = 3 > 0$ have the same sign, the object is speeding up.

10. $s(t) = t^3 - 3t^2 - 9t$

$$v(t) = 3t^2 - 6t - 9 \quad a(t) = 6t - 6$$

At $t = 4$: $v(4) = 3(16) - 6(4) - 9 = 48 - 24 - 9 = 15$. Speed = $|15| = 15$.

$$a(4) = 6(4) - 6 = 24 - 6 = 18.$$

Since $v(4) = 15 > 0$, the object is moving forward, with speed 15.

Since $v(4) > 0$ and $a(4) > 0$ have the same sign, the object is speeding up.

11. $s(t) = 16 - t^2$

$$v(t) = -2t \quad a(t) = -2$$

At $t = 1$: $v(1) = -2(1) = -2$. $a(1) = -2$.

Since $v(1) = -2 < 0$, the object is moving in the negative direction.

Since $v(1) < 0$ and $a(1) < 0$ have the same sign, the object is speeding up.

12. $s(t) = t^3 - 12t$

$$v(t) = 3t^2 - 12 \quad a(t) = 6t$$

At $t = 1$: $v(1) = 3(1) - 12 = -9$. $a(1) = 6(1) = 6$.

$v(1) < 0$ and $a(1) > 0$: opposite signs — slowing down (moving backward, but decelerating).

At $t = 3$: $v(3) = 3(9) - 12 = 15$. $a(3) = 6(3) = 18$.

$v(3) > 0$ and $a(3) > 0$: same signs — speeding up (moving forward, and accelerating).

Comparison: at $t = 1$ the object is moving backward and slowing down; at $t = 3$ it is moving forward and speeding up.

Challenge

13. $s(t) = t^3 - 6t^2 + 9t$

$$v(t) = 3t^2 - 12t + 9 = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3)$$

Setting $v(t) = 0$ gives $t = 1$ and $t = 3$ — these are the moments the object is momentarily at rest.

$$a(t) = 6t - 12$$

Around $t = 1$:

Just before ($t = 0.5$, from Problem 7): $v = 3.75 > 0$, $a = -9 < 0$ — opposite signs, slowing down, moving forward, approaching rest.

Just after ($t = 1.5$): $v(1.5) = 3(2.25) - 12(1.5) + 9 = 6.75 - 18 + 9 = -2.25$. $a(1.5) = 6(1.5) - 12 = -3$. Both negative — same signs, speeding up, now moving backward.

So at $t = 1$: the object stops, reverses direction, and starts speeding up moving backward.

Around $t = 3$:

Just before ($t = 2.5$): $v(2.5) = 3(6.25) - 12(2.5) + 9 = 18.75 - 30 + 9 = -2.25$. $a(2.5) = 6(2.5) - 12 = 3$. Opposite signs (v negative, a positive) — slowing down, still moving backward, approaching rest.

Just after ($t = 3.5$): $v(3.5) = 3(12.25) - 12(3.5) + 9 = 36.75 - 42 + 9 = 3.75$. $a(3.5) = 6(3.5) - 12 = 9$. Both positive — same signs, speeding up, now moving forward.

So at $t = 3$: the object stops again, reverses direction once more, and speeds up moving forward.

Full story: the object moves forward while slowing down, stops at $t = 1$, reverses and speeds up moving backward, then slows down again while still moving backward, stops at $t = 3$, and finally reverses again and speeds up moving forward.

14. $s(t) = 2t^3 - 3t^2 - 12t + 5$

$$v(t) = 6t^2 - 6t - 12 = 6(t^2 - t - 2) = 6(t - 2)(t + 1)$$

Setting $v(t) = 0$ gives $t = 2$ and $t = -1$.

Since we generally require $t \geq 0$ for a physical motion problem, only $t = 2$ is physically relevant ($t = -1$ is a valid mathematical solution but represents a time before the motion starts, so we set it aside.)

$$a(t) = 12t - 6$$

Around $t = 2$:

Just before ($t = 1.5$): $v(1.5) = 6(2.25) - 6(1.5) - 12 = 13.5 - 9 - 12 = -7.5$. $a(1.5) = 12(1.5) - 6 = 12$. Opposite signs (v negative, a positive) — slowing down, moving backward, approaching rest.

Just after ($t = 2.5$): $v(2.5) = 6(6.25) - 6(2.5) - 12 = 37.5 - 15 - 12 = 10.5$. $a(2.5) = 12(2.5) - 6 = 24$. Same signs (both positive) — speeding up, now moving forward.

So at $t = 2$: the object transitions from moving backward while slowing down, to moving forward while speeding up.

15. $s(t) = \sqrt{t} + t = t^{1/2} + t$

$$v(t) = \frac{1}{2}t^{-1/2} + 1 \quad a(t) = -\frac{1}{4}t^{-3/2}$$

At $t = 4$: $v(4) = \frac{1}{2}(4)^{-1/2} + 1 = \frac{1}{2} \cdot \frac{1}{2} + 1 = \frac{1}{4} + 1 = \frac{5}{4}$.

$$a(4) = -\frac{1}{4}(4)^{-3/2} = -\frac{1}{4} \cdot \frac{1}{8} = -\frac{1}{32} \text{ (since } 4^{3/2} = (\sqrt{4})^3 = 2^3 = 8\text{)}.$$

$$v(4) = \frac{5}{4}, \quad a(4) = -\frac{1}{32}.$$

16. $s(t) = \frac{1}{t+1}$

Using the quotient rule with $f = 1$ ($f' = 0$) and $g = t + 1$ ($g' = 1$):

$$v(t) = \frac{0 \cdot (t+1) - 1 \cdot 1}{(t+1)^2} = \frac{-1}{(t+1)^2}$$

For the acceleration, write $v(t) = \frac{-1}{(t+1)^2}$ and apply the quotient rule again, with $f = -1$ ($f' = 0$) and $g = (t+1)^2 = t^2 + 2t + 1$ ($g' = 2t + 2$):

$$a(t) = \frac{0 \cdot (t+1)^2 - (-1)(2t+2)}{(t+1)^4} = \frac{2t+2}{(t+1)^4} = \frac{2(t+1)}{(t+1)^4} = \frac{2}{(t+1)^3}$$

Sign of $v(t)$: since $(t+1)^2$ is always positive (it's a square) for $t \geq 0$, $v(t) = \frac{-1}{(t+1)^2}$ is always negative, for every single value of $t \geq 0$.

In words: this object's velocity is always negative and never equals zero, meaning it never comes to rest — it's always moving in the negative direction (always decreasing), though as t grows very large, that motion slows to a near-standstill without ever fully stopping.

17. Given $v(t) = t^2 - 4t + 3 = (t-1)(t-3)$ on $[0, 4]$.

$$a(t) = v'(t) = 2t - 4$$

The velocity is zero at $t = 1$ and $t = 3$, splitting $[0, 4]$ into subintervals: $[0, 1)$, $(1, 3)$, and $(3, 4]$.

On $[0, 1)$ (test $t = 0$): $v(0) = 3 > 0$ (moving forward). $a(0) = 2(0) - 4 = -4 < 0$. Opposite signs — slowing down, approaching rest at $t = 1$.

At $t = 1$: $v(1) = 0$ — momentarily at rest.

On $(1, 3)$, first half (test $t = 1.5$): $v(1.5) = 2.25 - 6 + 3 = -0.75 < 0$ (moving backward). $a(1.5) = 2(1.5) - 4 = -1 < 0$. Same signs — speeding up, moving backward.

On $(1, 3)$, second half (test $t = 2.5$): $v(2.5) = 6.25 - 10 + 3 = -0.75 < 0$ (still moving backward). $a(2.5) = 2(2.5) - 4 = 1 > 0$. Opposite signs — slowing down, still moving backward, approaching rest at $t = 3$.

(Right at $t = 2$, $a(2) = 0$ exactly — this is the instant the acceleration itself momentarily switches sign, marking the changeover from speeding up to slowing down within the backward motion.)

At $t = 3$: $v(3) = 0$ — momentarily at rest again.

On $(3, 4]$ (test $t = 4$): $v(4) = 16 - 16 + 3 = 3 > 0$ (now moving forward). $a(4) = 2(4) - 4 = 4 > 0$. Same signs — speeding up, moving forward.

Full story on $[0, 4]$: the object moves forward while slowing down, stops at $t = 1$, moves backward while speeding up, continues backward but starts slowing down around $t = 2$, stops again at $t = 3$, then moves forward while speeding up through $t = 4$.

Applied

18. $h(t) = -16t^2 + 200t + 10$

$$v(t) = -32t + 200 \quad a(t) = -32$$

$v(3) = -32(3) + 200 = -96 + 200 = 104$. $a(3) = -32$ (always -32 , representing constant gravity).

Since $v(3) = 104 > 0$, the rocket is still moving upward at $t = 3$ seconds.

Since $v(3) > 0$ and $a(3) < 0$ have opposite signs, the rocket is slowing down — it's still rising, but gravity is steadily pulling it back, decelerating its ascent as it climbs toward its peak height.

19. $y(t) = t^3 - 9t^2 + 24t + 2$

$$v(t) = 3t^2 - 18t + 24 \quad a(t) = 6t - 18$$

At $t = 1$: $v(1) = 3 - 18 + 24 = 9$. $a(1) = 6 - 18 = -12$.

Since $v(1) = 9 > 0$, the elevator is moving upward at $t = 1$.

Since $v(1) > 0$ and $a(1) < 0$ have opposite signs, the elevator is slowing down — it's still rising, but decelerating at that moment (perhaps easing toward a stop at a floor).

20. $P(t) = t^3 - 5t^2 + 3t + 60$

$$P'(t) = 3t^2 - 10t + 3 \quad P''(t) = 6t - 10$$

At $t = 1$: $P'(1) = 3 - 10 + 3 = -4$. $P''(1) = 6 - 10 = -4$.

Since $P'(1) = -4 < 0$, the stock price is currently decreasing at $t = 1$ month.

Since $P'(1) < 0$ and $P''(1) < 0$ have the same sign, the price decline is accelerating — the stock isn't just losing value, it's losing value at a faster and faster rate at that moment.

21. $h(t) = 100 - 16t^2$

$$v(t) = -32t \quad a(t) = -32$$

$v(2) = -32(2) = -64$. Speed = $|-64| = 64$ ft/s.

$a(t) = -32$ for every value of t — it never changes.

In words: the acceleration is a fixed constant (-32 ft/s², matching the constant pull of gravity near Earth's surface) at every single moment of the fall — it doesn't depend on t at all. This tells you that gravity is applying the exact same downward acceleration throughout the entire fall, which is why $v(2) = -64$ ft/s represents the object continuously speeding up (in the negative/downward direction) the whole way down, right up until it hits the ground.